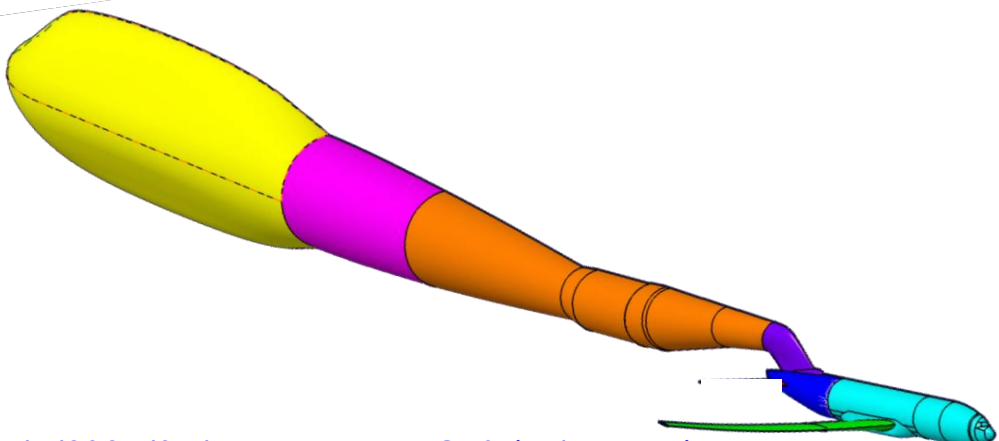


Test Case 2a: T&I Study (Wing/Body)

- CRM Wing/Body with Upper Swept Strut

1. Wing/Body (2.7% model scale in tunnel)
2. Wing/Body + Upper Swept Strut
 - Wing deformed to matching condition (from DPW7)
 - Aft strut surface (shown in yellow) replaces interface to arc sector



- Geometry

- https://commonresearchmodel.larc.nasa.gov/wp-content/uploads/sites/7/2025/07/DPW-7_WBT_IGES_in_low_q.zip
- [https://dpw.larc.nasa.gov/DPW8/Test Environment/Test Case 1/Sting No Arc Sector Bulb noRotation 2025 09 03.igs](https://dpw.larc.nasa.gov/DPW8/Test_Environment/Test_Case_1/Sting_No_Arc_Sector_Bulb_noRotation_2025_09_03.igs)

- Conditions & Reference Units

Mach	Re _c	T _{static} (120° F)	α
0.85	5 × 10 ⁶	579.67 R 322.04 K	−1.50°, 0.00°, 1.50°, 2.50°, 2.75°, 3.00°, 3.50°, 4.00°, 4.50°
Sref (semi-span grid)		Cref	Semispan
216.77544 sq.in		7.5195 in	31.23225 in
Moment Center			
(156.00, 0.00, 0.00)			

- Comparison metrics

- Forces / Moments
- Sectional C_p distribution
- Residuals (Flow & Structural Solver)

Comparison Data

NTF197: r44,r51,r53
NTF215: r43,r103
NTF229: r296,r300,r302
Ames216: r35,r126,r130,r133